MILITARY SPECIFICATION

SPRAY OUTFIT, PAINT; BITUMINOUS EMULSION, TRAILER-MOUNTED, WITH COMPRESSOR AND GASOLINE ENGINE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a complete trailer-mounted spray outfit for spraying bituminous emulsions and other liquids from a standard 55-gallon shipping drum.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-S-571	- Solder; Tin Alloy; Lead-Tin Alloy and Lead Alloy.
ZZ-H-461	- Hose, Rubber, Gas (Acetylene-Hydrogen, Air and Oxygen).
ZZ-H-521	- Hose; Spray.

Military

MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine (Heavy Duty).

FSC 4940

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MIL-G-3859	- Grease Guns, Hand, High Pressure, Lever-Operated; Cartridge and Bulk Loading (14 Ounce and 21 Ounce Capacity).
MIL-C-3925	- Compressors, Reciprocating and Rotary, Power Driven, Air, (Paint Spray Type).
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-S-12651	- Sprayers, Packaging of.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-T-46786	- Tanks, Fuel, Engine, General Requirements for.

STANDARDS

Military

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U. S. Military Property.
MIL-STD-1226	- Engine, Gasoline: Air-Cooled, 1-1/2 BHP, 4-Cycle, Military Design, Model 1A08 Installation Procedures.
MS15003	- Fittings, Lubrication (Hydraulic) Surface Check, 1/8 Pipe Thread, Steel, Type III.
MS24207	- Coupling, Grease Gun; Hydraulic Type Nozzle.
MS35389	- Tire, Pneumatic: Industrial Straight Side, Tractor Rear and Tractor Front.
MS35392	- Inner Tube: Pneumatic Tire.
MS51336	- Lunnette - Coupler, Drawbar, Ring.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

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3. REQUIREMENTS

3.1 Description. The spray outfit shall be a complete, self-contained unit consisting essentially of a four-wheel, rubber-tired, towed-type trailer; gasoline engine; air compressor; material straining chamber; siphon tube; and two spray guns. The spray outfit shall spray paints, asphalt cutbacks, bituminous emulsions, weed killers, insecticides, and disinfectants directly from the shipping drum.

3.2 Preproduction model. The supplier shall furnish a spray outfit for examination and test within the time frame specified to prove, prior to starting production, that his production methods and choice of design detail will produce spray outfits that comply with the requirements of this specification (see 6.3). Examination and tests shall be those specified herein.

3.3 Material. Material shall be as specified herein. Material not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.4 Safety. All exposed parts which are subject to high operating temperatures shall be insulated, enclosed, or guarded. All moving parts which are of such nature or so located as to be a hazard to operating personnel shall be enclosed or guarded. Protective devices shall not impair the operating functions.

3.5 Gasoline engine. The gasoline engine shall be a 1-1/2 horsepower (hp) Military Standard, Model 1A08. The engine shall be furnished by the Government and installed by the supplier in accordance with the procedures outlined in MIL-STD-1226 (see 6.4).

3.5.1 Fuel tank. The supplier shall furnish and install a fuel tank with a capacity for not less than 4 hours operation. The fuel tank shall be made of terne-plated material in accordance with MIL-T-46786, type I. The seams and joints shall be soldered with solder conforming to QQ-S-571, composition Sn 30-70. The construction and mounting of the tank shall be such that vibration, excessive tightening of the holddown devices, or contact with other parts of the engine will not cause leaks to develop.

3.6 Air compressor. The air compressor shall conform to MIL-C-3925, type I, style B, group a, size 1 and shall be V-belt driven. A belt guard shall be provided.

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3.7 Trailer. The trailer shall withstand towing over unsurfaced roads at speeds of not less than 10 mph without permanent deformation, malfunction, or failure of any part or component.

3.7.1 Frame. The trailer frame shall be of all metal welded construction and shall be provided with a metal towbar. The towbar shall control the steering of the front wheels and shall be equipped with a lunette ring conforming to MS51336. The frame shall support a full 55-gallon drum in a horizontal position. Reinforcements shall be provided to prevent failure of the drum heads when subjected to pressures above 35 psig. The engine and compressor assembly shall be mounted on the frame.

3.7.2 Suspension. The trailer shall be equipped with four pneumatic tired wheels. The wheels shall be mounted on adjustable antifriction bearings. The tires shall be tube type conforming to MS35389-3. The tubes shall conform to MS35392-50.

3.7.3 Spray hose brackets. Brackets to carry the lengths of hose specified herein shall be attached to the trailer in a location to prevent damage to the hose while in transit.

3.8 Hose.

3.8.1 Material. Two 20-foot lengths of 3/4-inch diameter hose shall be furnished. The hoses shall conform to ZZ-H-521, type II, grade A. The hoses shall terminate with standard pipe threads and packless hand unions.

3.8.2 Air. One 5-foot length of 3/8-inch diameter air hose conforming to ZZ-H-461, type II, style A shall be furnished. The necessary fittings for attaching the hose to the air compressor shall be supplied. The hose shall attach to the discharge tube assembly with a packless hand union.

3.9 Discharge tube assembly. The discharge tube assembly shall incorporate the air inlet line, the material discharge tube, a bung adapter for standard bungs and the necessary piping and fittings to connect the discharge tube to the material strainer. All fittings, adapters, and connections shall incorporate packless hand unions.

3.10 Strainer. The strainer shall be the quick-opening type with a removable screen. The screen shall have an area of not less than 50 square inches with perforations sized to pass only the materials specified in 3.1. The screen shall withstand without damage the burning of paint in the process of cleaning.

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3.11 Pressure-flushing attachment. A pressure-flushing attachment with a tank of not less than 1/2-gallon capacity shall be furnished with the unit. The tank shall be located so that insertion of the discharge tube will result in the cleaning solvent being forced through the spray heads.

3.12 Spray pipes. The spray outfit shall be equipped with two spray pipes not less than 4-1/2 feet long. The spray pipes shall be double offset with manual shutoff cocks. The spray pipes shall be threaded on each end to receive the hose and the nozzle.

3.13 Nozzles. The nozzles provided with each spray gun shall be suitable for the following methods of application.

3.13.1 Sheet application. A wide soft sheet application of emulsions or light cutbacks at approximately 6 gpm, where a coverage of 1/6 gallon per square yard or heavier is desired.

3.13.2 Diffused spray. A diffused spray, at approximately 4 gpm, for coating extensive areas with 1/10 gallon per square yard or heavier.

3.13.3 Conical spray. A conical spray, at approximately 4 gpm, for spraying asphalt emulsion, cutback asphalt, and materials of similar viscosity at a rate of 1/10 gallon per square yard or heavier.

3.13.4 Fine spray. A fine diffusion at approximately 2-1/2 to 3 gpm, for applications requiring less than 1/10 gallon per square yard.

3.14 Accessories. Each spray outfit shall be furnished with a combination adapter and bung wrench, two strainer screens, and a canvas engine cover.

3.15 Threads. All threaded parts shall conform to Handbook H28.

3.16 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating.

3.16.1 Lubricants. The spray outfit shall operate as specified herein when lubricated with military lubricants.

3.16.2 Lubrication fittings. Lubrication fittings shall conform to MS15003. Fittings shall be located in a protected position and shall be accessible to a grease gun conforming to MIL-G-3859, equipped with a hydraulic coupler conforming to MS24207 with a 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts.

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3.16.3 Pressure-release device. A pressure-release device shall be provided where the use of 1,000 psi and higher pressure lubricating equipment will damage grease seals or other parts.

3.16.4 Enclosures with integral reservoir of lubricant. Enclosures such as gear cases and transmission housings which contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with dipsticks, check plugs not less than 1/2-inch pipe size, or sight gages to determine the level of the lubricant. Each enclosure shall be equipped with a means for filling the enclosure with lubricant and for draining. The drain shall be fitted with a drain plug. The drain shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Drainage shall be to the ground when the spray outfit is in its normal position. Integral tubes or troughs may be used to convey the lubricant from the drain to the ground. Accessibility to the drain plug, the filling means, and the lubricant-level checking device shall be provided without the removal or adjustment of accessories or parts.

3.16.5 Lubrication service.

3.16.5.1 Grease lubrication. Bearing lubrication, including sealed bearings, shall be with greases as follows:

(a) MIL-G-23827 - For instrument and all sealed bearings.
(b) MIL-G-10924 - For other bearings where operating temperatures do not exceed 175° F.

The spray outfit shall be run-in (broken-in) and delivered with these greases. The bearings shall be cleaned before lubrication, because military greases may not be compatible with other greases. Each outfit shall be tagged in a conspicuous place to indicate which military grease has been used.

3.16.5.2 Oil lubrication. Unless otherwise preserved for packaging (see 5.1), all components shall be filled to the operating level with winter grades of oil conforming to MIL-L-2104 designated for use in the temperature range of 0° F. to 32° F. A tag shall be attached in a conspicuous place to indicate the temperature ranges and grades of lubricants used.

3.17 Identification marking. The spray outfit shall be identified in accordance with MIL-STD-130. The marking shall be applied to the spray outfit on plates conforming to MIL-P-514, type I, style 2, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

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3.18 Treatment and painting. All parts of the spray outfit normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A.

3.19 Technical publications. Operating instructions and such technical publications as are specified shall be furnished.

3.20 Repair parts and maintenance tools. Such repair parts and maintenance tools as are specified shall be furnished.

3.21 Toolbox. A toolbox made of steel not less than 0.075 inch thick shall be attached to the spray outfit. The toolbox shall be of a size to hold the operating and maintenance tools and repair parts specified. The toolbox shall be securely mounted in an accessible, protected location permitting the lid to be opened through an arc of 90 degrees. The lid shall remain in the fully open position by the weight of the lid or by holding devices. Provisions shall be made to drain the toolbox in its mounted position.

3.22 Items to be furnished with spray outfit. No shipment of spray outfits shall be made unless repair parts, maintenance tools, operating tools, and accessories are included with the shipment or unless approval for shipment of the spray outfits without such items has been received from the contracting officer.

3.23 Workmanship. All parts, components, and assemblies of the spray outfits including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux and other harmful, extraneous material. External surfaces shall be free from burrs, sharp edges, and corners except when sharp edges and corners are required.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction spray outfit shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The preproduction spray outfit shall be tested as specified in 4.5.2.1 through 4.5.2.4. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each spray outfit shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Each spray outfit shall be tested as specified in 4.5.2.1. Failure of a test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The spray outfits shall be examined for the following defects:

101. Engine installation not as specified.
102. Fuel tank not as specified.
103. Components and accessories missing or not as specified.
104. Identification marking missing or illegible.
105. Treatment and painting not as specified.
106. Technical publications missing or not as specified.
107. Repair parts and maintenance tools missing or not as specified.
108. Workmanship not as specified.

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4.5.2 Tests.

4.5.2.1 Operational. Start the engine and operate the spray outfit for 15 minutes with the spray system pressurized at not less than 35 psig. Inability of the compressor to develop 35 psig, leakage, or rough and erratic operation shall constitute failure of this test.

4.5.2.2 Road. Tow the spray outfit, with a full 55-gallon drum secured in the drum support and ready for operation, over unsurfaced roads for 2 miles at speeds of not less than 20 mph. Wheel bearing failure, permanent deformation, malfunction, or breakage of any part or component shall constitute failure of this test.

4.5.2.3 Performance. Operate the spray outfit until 25 gallons of spray material has been forced through the system. Test each nozzle for conformance to spray patterns and flow rate. Inability to spray 25 gallons of material; air or fluid leakage; inadequate flow rates; defective, split or one sided spray patterns shall constitute failure of this test.

4.5.2.4 Flushing attachment. At the completion of the performance test, flush the hose lines, gun, and nozzle with the pressure-flushing attachment. Inability to purge the lines, gun, and nozzle of spray material shall constitute failure of this test.

4.6 Inspection of preparation for delivery. The preservation, packaging, packing, and marking shall be examined and tested to determine compliance with the referenced documents and additional requirements in section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Each complete sprayer with repair parts, tools, publications, and accessories shall be preserved, packaged, and packed in accordance with MIL-S-12651. The level of preservation, packaging, and packing shall be level A, B, or C as specified. In addition, the engine cover shall be placed and secured on the engine, and when level A or B packing is specified, the sprayers shall be packed as specified for boxed sprayers.

5.2 Marking. In addition to any special marking required by the contract or order, packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The spray outfit is intended to be used in the spraying of paints, asphalt cutbacks, emulsions, weed killers, and disinfectants.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of the preproduction model (see 3.2).
(c) Operating instructions and technical publications required (see 3.19).
(d) Repair parts and maintenance tools required (see 3.20).
(e) Level of preservation and packaging and level of packing required (see 5.1).

6.3 Preproduction model. Any changes or deviations of production spray outfits from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish spray outfits conforming to this specification.

6.4 The contracting officer should arrange to furnish the property specified in 3.5.

Custodians:

Army - ME
Navy - YD
Air Force - 82

User activity:

Army - CE
Navy - MC

Preparing activity:

Army - ME

Project No. 4940-0178

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